

Network Security

Theory Class Nº3

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Outline

- Cryptography
  - Introduction
  - Symmetric cryptography



- Perspective:
  - User
  - Manager
  - Developer

Introduction to Cryptography



- Cryptography is the science of using mathematics to encrypt and decrypt data

- Cryptography is important:
  - Data integrity
  - User authentication
  - Information confidentiality

"A good cryptographic algorithm must be based on a balanced compromise between complexity and usability"

Introduction to Cryptography

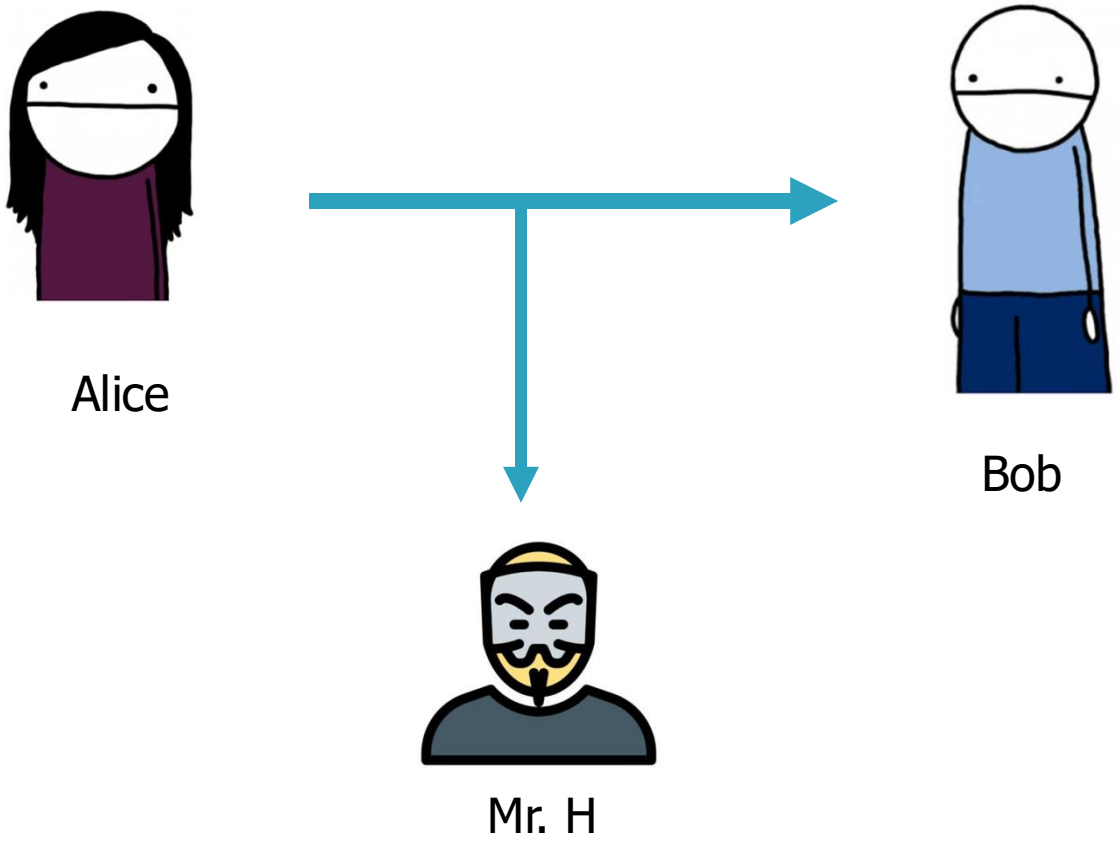
- Bank cards
- Electronic commerce
- Mobile phones
- Internet
- Emails
- Blockchain
- DarkWeb
- ...

Cryptography History

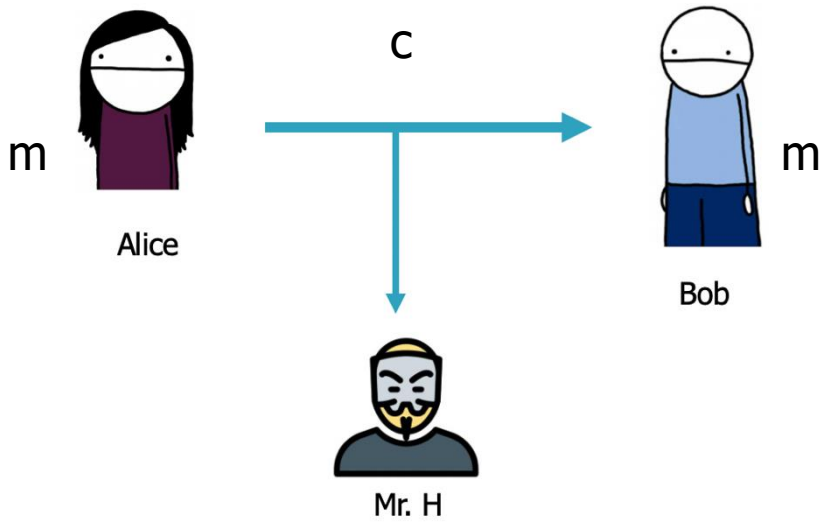
- Cryptography derives its name from a Greek word “Kryptos” which means “Hidden Secrets”.

- Science
  - Maths
  - Computer Science

Cryptography



Cryptography (confidentiality)



m – Plain text
k – secret key
E – encryption algorithm
D – decryption algorithm
c – Cypher text

c = E (k , m)
m = D (k, c)
D (k , E (k , m)) = m

Cryptography



"A security system is as strong as its weakest component"

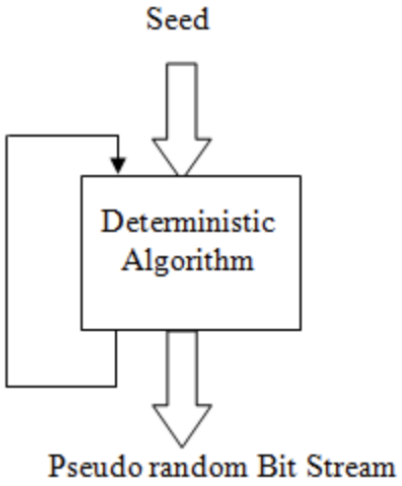
Kerckhoffs's principle

- The security of the encryption scheme must depend only on secrecy of the key and not on the secrecy of the algorithm
- As opposed to "security through obscurity"
- New ciphers and algorithms are not trusted without public scrutiny over years.



Some concepts

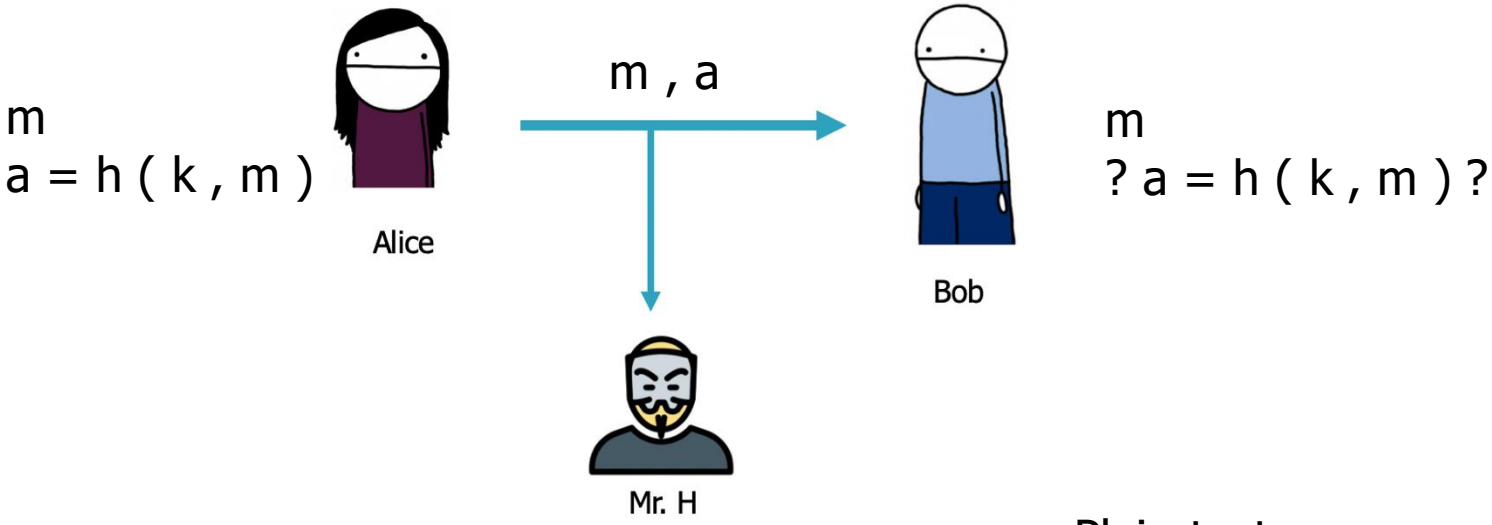
- One-way (irreversible) functions
  - Difficult to apply y=f(x)
  - Ex: m mod n = z
- Random and pseudo-random number generators
  - Problem: period
    - May not be a problem if the seed is not known and the period is large enough
  - Need to restart with (random) seeds



Encryption Strength

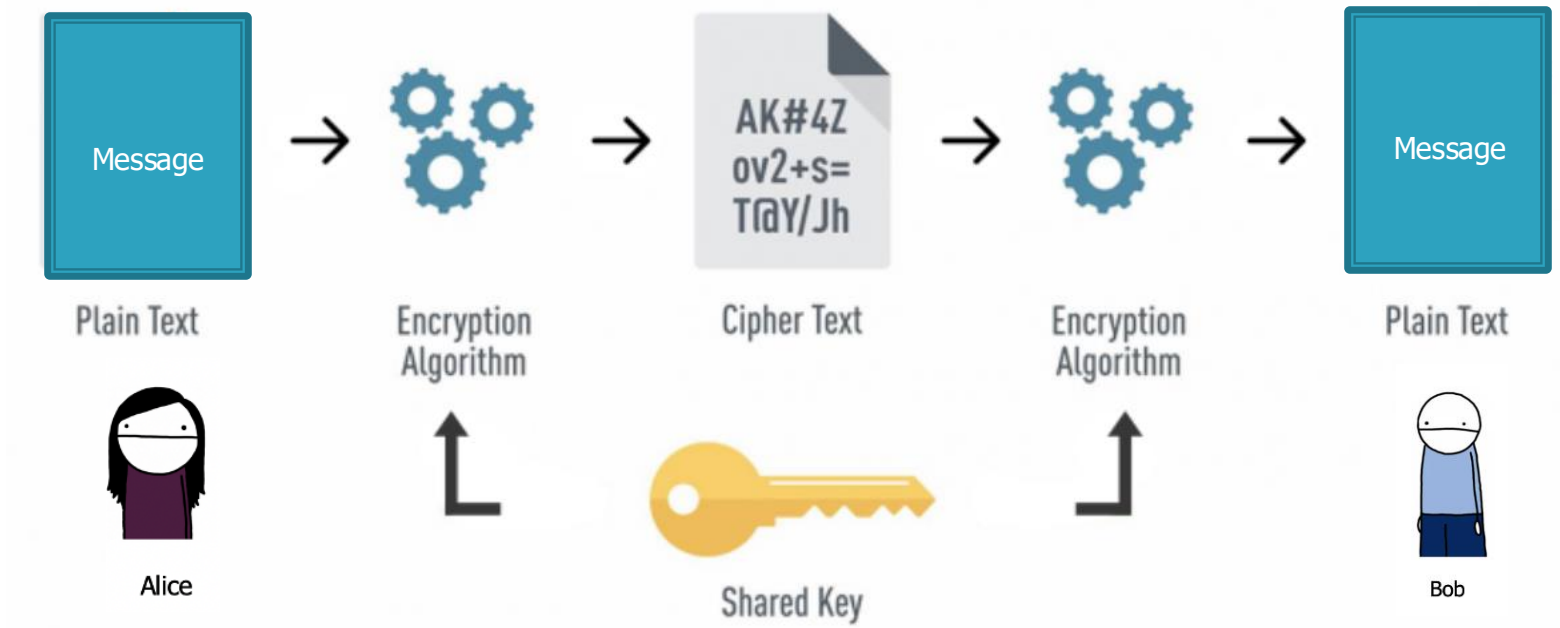
- Depends on:
  - Algorithm
  - The secrecy of the key
  - The length of the key
  - The initialization vectors
  - How all work together
- Work factor
  - An estimate of the effort and resources it would take to penetrate a cryptosystem
- Strength of protection mechanism should be based in correlation to the sensitivity of the data being encrypted.

Authentication



m – Plain text
k – Secret key
h – Cryptographic function for authentication

Symmetric Encryption

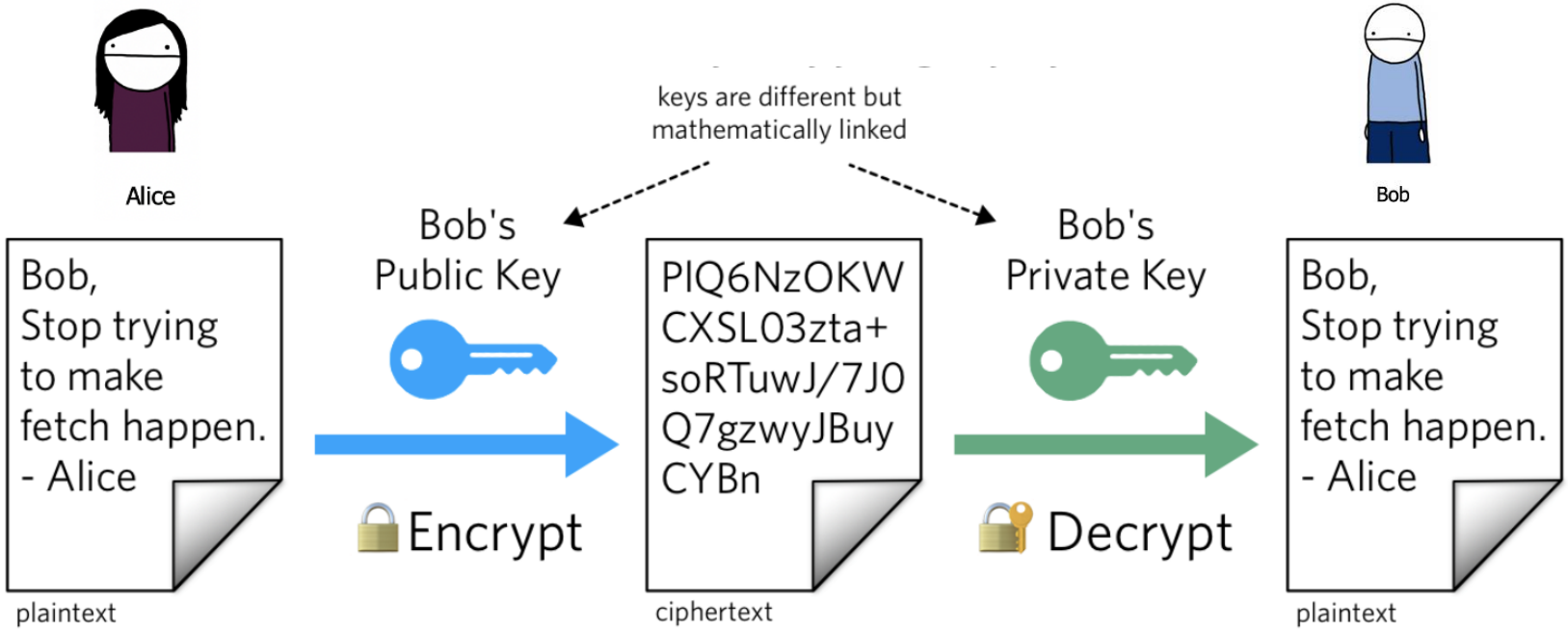


Symmetric Encryption

- Requirements for secure use of symmetric encryption:
  1. Strong encryption algorithm: Given the algorithm and ciphertext, an attacker cannot obtain key or plaintext
  2. Sender/receiver know secret key (and keep it secret)

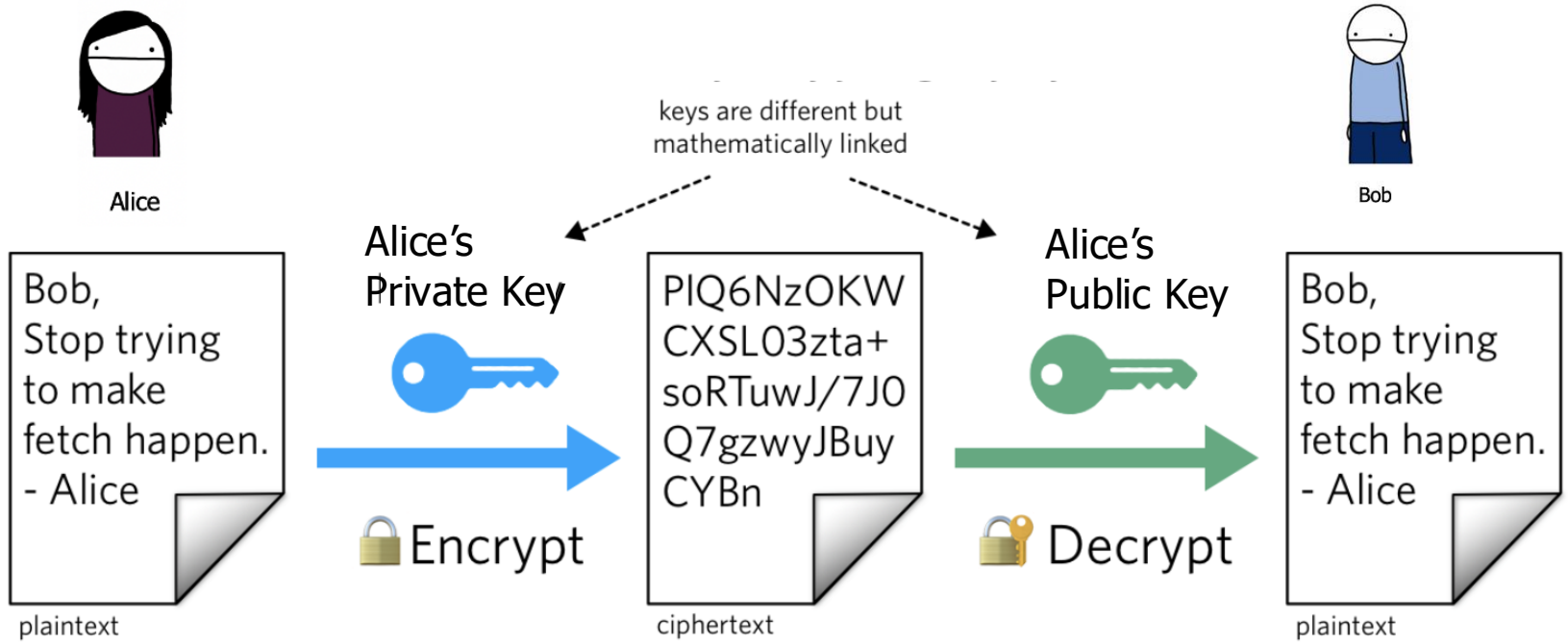
- Assumptions:
  - Cipher is known
  - Secure channel to distribute keys

Asymmetric Encryption (confidentiality)



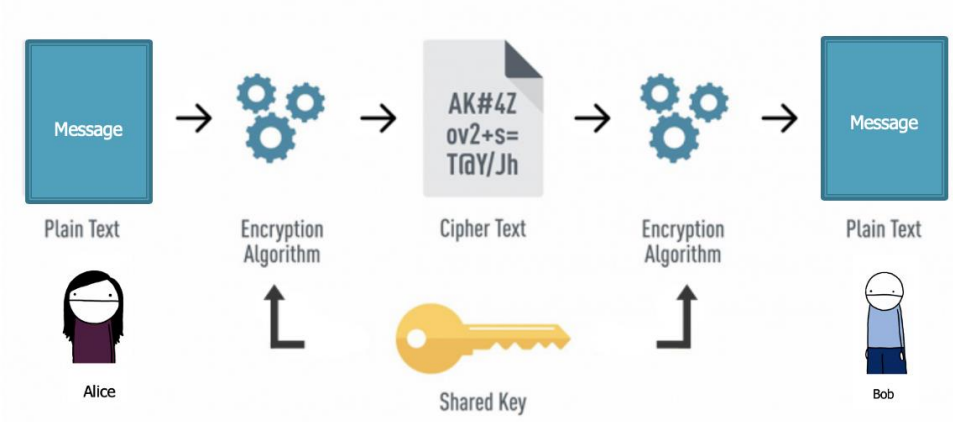
Only Bob can read the message!

Asymmetric Encryption (authentication)



Symmetric Encryption

- Less computational expensive than asymmetric encryption
- Challenge: the distribution of the key
- Usually is used combined with asymmetric encryption



Symmetric vs Asymmetric

- Asymmetric encryption requires more computational power
- Asymmetric encryption can be used to distribute symmetric keys
- Symmetric keys should be reset from time to time

Characterising Cryptographic Systems



- Operations used for encryption:
  - Substitution: replace one element in plaintext with another
  - Transposition: re-arrange elements
  - Product systems: multiple stages of substitutions and transpositions
- Number of keys used:
  - Symmetric: sender/receiver use same key (single-key, secret-key, shared-key, conventional)
  - Public-key: sender/receiver use different keys (asymmetric)
- Processing of plaintext:
  - Block cipher process one block of elements at a time
  - Stream cipher process input elements continuously

Cryptanalysis and Brute – Force Attacks

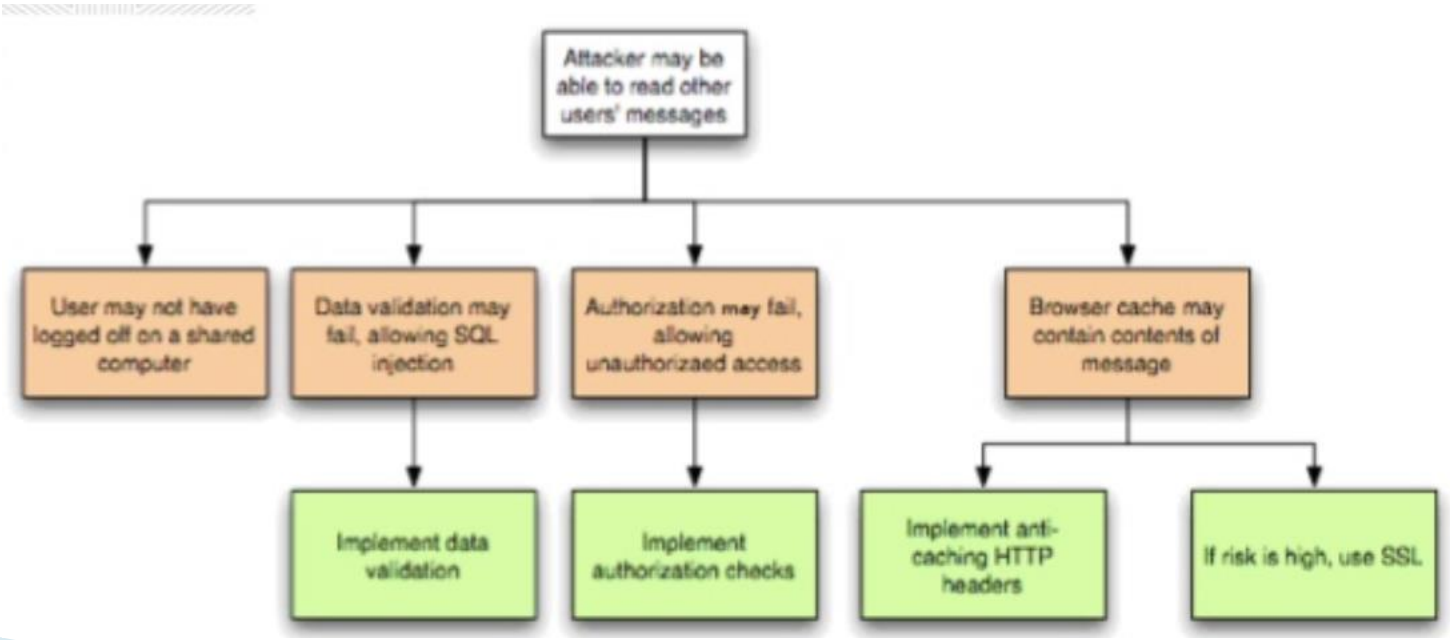
- Objective of attacker: recover key (not just message)
- Approaches of attacker:
  - Cryptanalysis: Exploit characteristics of algorithm to deduce plaintext or key
  - Brute-force attack: Try every possible key on ciphertext until intelligible translation into plaintext obtained
- Important note: if attack succeeds, all future/past messages are compromised

Type of attacks*

- Generic (Black Box) vs Specific
- Passive vs Active
  - Active
    - Masquerading
    - Replay
    - Message changing
    - Denial of Service
- Attacks based on text
  - Ciphertext-only attacks
  - **Known plaintext attacks** (ex: automatic email)
  - Chosen plaintext attacks
  - Chosen ciphertext attacks
- Collision attacks
  - Birthday attack
    - For N values there is a high probability of duplication after root(N)
  - Meet-in-the-middle attack
    - Use of tables with different keys precalculated.

Attack Trees

- Hierarchical, graphical diagrams that show how low level hostile activities interact and combine to achieve an adversary's objectives
- Important also to the owners





Symmetric Encryption

Classic techniques

(Before computer age…)

- Substitution techniques
- Transposing techniques
- Rotor machines
- Steganography

Substitution techniques

- Letters of plaintext are replaced by others letters or by numbers of symbols
- Examples:

1. Caesar Cipher

- Earliest known cipher, used by Julius Caesar (Roman general 2000 years ago)
- Replace each letter by the letter three positions along in alphabet

Plain : a b c d e f g h i j k l m n o p q r s t u v w x y z
Cipher: D E F G H I J K L M N O P Q R S T U V W X Y Z A B C

2. Generalised Caesar Cipher

- Allow shift by k positions
- Assume each letter assigned number (a = 0, b = 1, . . .)

  $C = E(k; p) = (p + k) \bmod 26$

  $p = D(k; C) = (C - k) \bmod 26$

Substitution techniques

3. Atbash cypher

- ”The Da Vinci code”

a b c d e f g h i j k l m
z y x w v u t s r q p o n

4. Monoalphabetic cypher

- Use a single alphabet for both plaintext and ciphertext
- Arbitrary substitution: one element maps to any other element
- n element alphabet allows n! permutations or keys
- Example:

Plain : a b c d e ... w x y z
Cipher: D Z G L S ... B T F Q

-> Theoretical: 26!
-> Real: << 26!

- Regularities of the language
- Frequency of letters, diagrams, trigrams
- Expected words

Substitution techniques

4. Polyalphabetic cypher

- Use a set of characters for both plaintext and ciphertext
- Example:

Plain : ab ac ad ae ...
Cipher: ND UZ PO LI ...

5. Vigenère cypher

- Set of 26 general Caesar ciphers
- Letter in key determines the Caesar cipher to use
- Key must be as long as plaintext: repeat a keyword
- Example:

Plain: internettechnologies
Key: sirindhornsirindhorn
Cipher: AVKMEQLHKRUPEWYRNWVF

- Multiple ciphertext letters for each plaintext letter

Substitution techniques

6. One-time pad

- Similar to Vigenere, but use random key as long as plaintext
- Perfect secrecy! – only known scheme that is unbreakable (unconditional security)
- Ciphertext has no statistical relationship with plaintext
- Given two potential plaintext messages, attacker cannot identify the correct message
- Two practical limitations:
  - Difficult to provide large number of random keys
  - Distributing unique long random keys is difficult
- Limited practical use

One-time pad

ENCRYPT

	0 0 1 1 0 1 0 1	Plaintext
⊕	1 1 1 0 0 0 1 1	Secret Key
=	1 1 0 1 0 1 1 0	Ciphertext

DECRYPT

	1 1 0 1 0 1 1 0	Ciphertext
⊕	1 1 1 0 0 0 1 1	Secret Key
=	0 0 1 1 0 1 0 1	Plaintext

Transposition techniques*

1. Rows/Columns Transposition

- Plaintext letters written in rows
- Ciphertext obtained by reading column-by-column, but re-arranged
- Key determines order of columns to read
- Easy to break using letter frequency (try different column orders)
- Transposition ciphers can be made stronger by using multiple stages of transposition

Transposition techniques

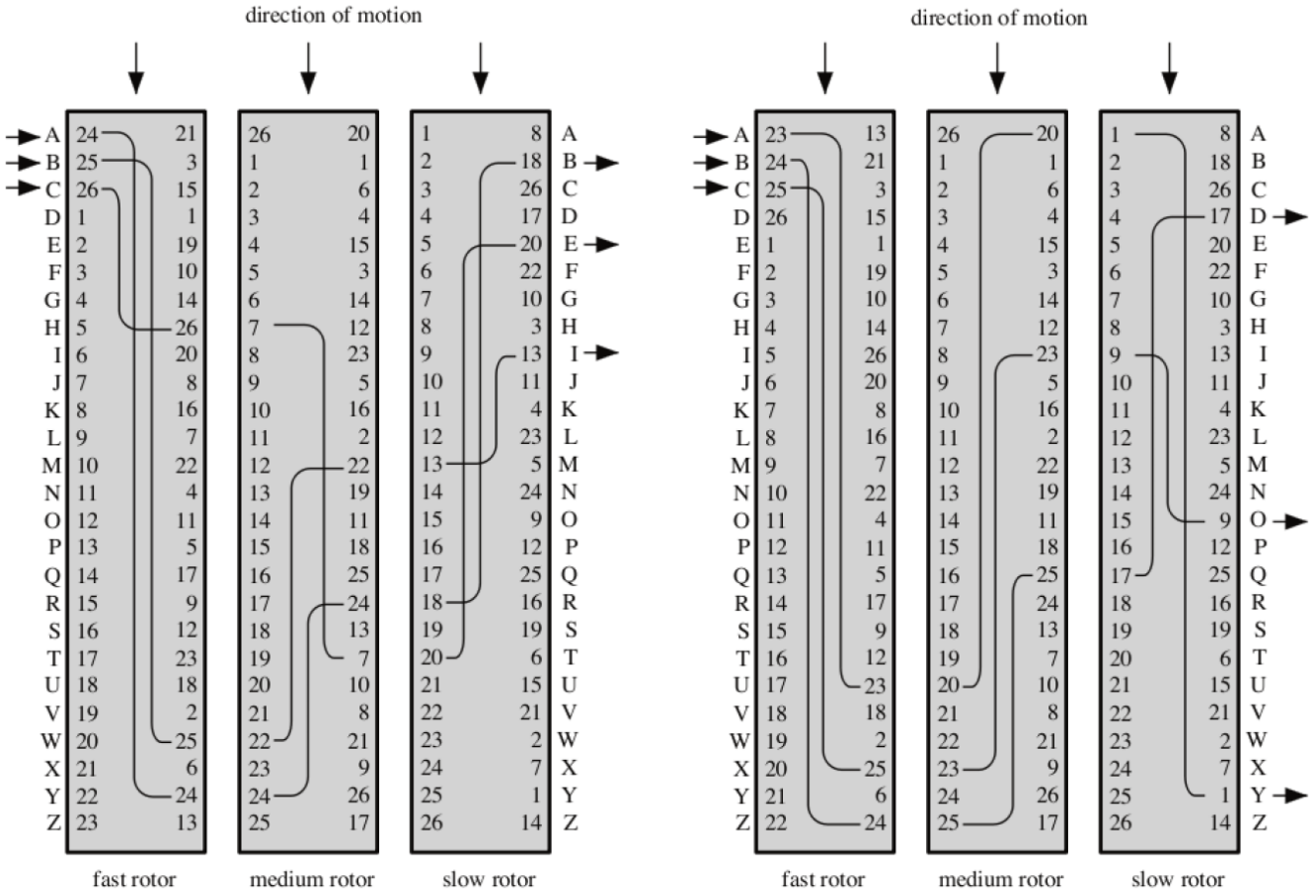
1. Rows/Columns Transposition

```
Key:        4 3 1 2 5 6 7
Input:      t t n a a p t
            m t s u o a o
            d w c o i x k
            n l y p e t z
Output:     NSCYAUOPTTWLTMDNAOIEPAXTTOKZ
```

Rotor machines

- Multiple stages of encryption can be used for substitution and transposition ciphers
- Rotor machines were early application of this principle was basis for Enigma cipher used by Germany in WW2
- Machine has multiple cylinders
- Monoalphabetic substitution cipher for each cylinder
- Output of one cylinder is input to next cylinder
- Plaintext is input to 1rst cylinder; ciphertext is output of last cylinder
- Entering a plaintext letter causes last cylinder to rotate its cipher
- Complete rotation of one cylinder causes previous cylinder to rotate its cipher
- Principle is used in Data Encryption Standard (DES)

Rotor machines



Steganography

- Hide a real message in a fake, but meaningful, message
- Assumes recipient knows the method of hiding
- Examples:
  - Selected letters in a document are marked to form the hidden message
  - Invisible ink (letters only become visible when exposed to a chemical or heat)
  - Using selected bits in images or videos to carry the message
- Advantages
  - Does not look like you are hiding anything
- Disadvantages
  - Once attacker knows your method, everything is lost
  - Can be inefficient (need to send lot of information to carry small message)

Measures of Security

- Unconditionally Secure
  - Ciphertext does not contained enough information to derive plaintext or key
  - One-time pad is only unconditionally secure cipher (but not very practical)

- Computationally Secure
  - If either:
    - Cost of breaking cipher exceeds value of encrypted information
    - Time required to break cipher exceeds useful lifetime of encrypted information
  - Hard to estimate value/lifetime of some information
  - Hard to estimate how much effort needed to break cipher

Examples of Brute-Force Attacks

Key Size (bits)	Number of Alternative Keys	Time Required at 1 Decryption/μs	Time Required at 10^6 Decryptions/μs
32	$2^{32} = 4.3 \times 10^9$	$2^{31}\ \mu s$ = 35.8 minutes	2.15 milliseconds
56	$2^{56} = 7.2 \times 10^{16}$	$2^{55}\ \mu s$ = 1142 years	10.01 hours
128	$2^{128} = 3.4 \times 10^{38}$	$2^{127}\ \mu s$ = 5.4×10^{24} years	5.4×10^{18} years
168	$2^{168} = 3.7 \times 10^{50}$	$2^{167}\ \mu s$ = 5.9×10^{36} years	5.9×10^{30} years
26 characters (permutation)	$26! = 4 \times 10^{26}$	$2 \times 10^{26}\ \mu s = 6.4 \times 10^{12}$ years	6.4×10^6 years



Questions?